Cluster-Memory Augmented Deep Autoencoder via Optimal Transportation for Hyperspectral Anomaly Detection

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***Abstract*— Hyperspectral anomaly detection (AD) aims to detect objects significantly different from their surrounding background. Recently, many detectors based on autoencoder (AE) exhibited promising performances in hyperspectral AD tasks. However, the fundamental hypothesis of the AE-based detector that anomaly is more challenging to be reconstructed than background may not always be true in practice. We demonstrate that an AE could well reconstruct anomalies even without anomalies for training, because AE models mainly focus on the quality of sample reconstruction and do not care if the encoded features solely represent the background rather than anomalies. If more information is preserved than needed to reconstruct the background, the anomalies will be well reconstructed. This article proposes a cluster-memory augmented deep autoencoder via optimal transportation for hyperspectral anomaly detection (OTCMA) clustering for hyperspectral AD to solve this problem. The deep clustering method based on optimal transportation (OT) is proposed to enhance the features consistency of samples within the same categories and features discrimination of samples in different categories. The memory module stores the background's consistent features, which are the cluster centers for each category background. We retrieve more consistent features from the memory module instead of reconstructing a sample utilizing its own encoded features. The network focuses more on consistent feature reconstruction by training AE with a memory module. This effectively restricts the reconstruction ability of AE and prevents reconstructing anomalies. Extensive experiments on the benchmark datasets demonstrate that our proposed OTCMA achieves state-of-the-art results. Besides, this article presents further discussions about the effectiveness of our proposed memory module and different criteria for better AD.**

***Index Terms*— Anomaly detection (AD), deep clustering, hyperspectral images, memory module, optimal transportation (OT).**

I. Introduction

ANOMALY detection (AD) is to automatically discern the anomaly object that is significantly different from the normal objects [1], [2]. In hyperspectral images (HSIs), anomaly could be an interesting object of land cover [3], a dangerous event in sea rescue and daily monitoring [4], [5], and the abnormal crop growth in crop monitoring [6], [7]. Benefiting from fine-grained spectral characteristics, the application of HSI AD has shown great potential in many different fields. However, there are numerous challenges. First, the labels of anomaly and background are always unavailable because of the heavy labor cost and lacking professional area knowledge. Second, the anomaly is unpredictable and the minority, so it is impossible to model the anomaly and cover all the cases sufficiently. Third, the background of HSI is sometimes very complex and heterogeneous in piratical, increasing the false alarm rate of the detection results. For better AD, deep models based on sample reconstruction are widely studied to facilitate detection performance. As one of the typical reconstruction-based deep models, autoencoder (AE) is utilized to build the detector by assuming that anomalies are the minority and are insufficient to train AE to be well reconstructed while the background can. As shown in Fig. 1, pixels with larger reconstruction errors (closer to red) are more likely to be anomalies. However, we find that this assumption may not always be true. As presented in Fig. 2(a) and (b), even if no anomalies train the convolutional AE (CAE), the anomalies can be still reconstructed. The reason can be concluded as follows. First, there is no constraint to ensure that the features of one class's samples can only represent themselves, resulting in more information being retained, which may contribute

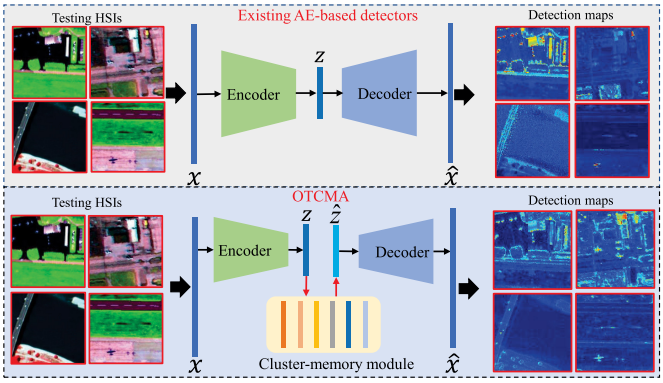


Fig. 1. Existing AE-based detectors reconstruct a testing pixel by its own feature z outputting from the encoder. In contrast, our proposed OTCMA first inputs the encode z into the cluster-based memory module and then generates the consistent feature $\hat{z}$ for the subsequent pixel reconstruction.

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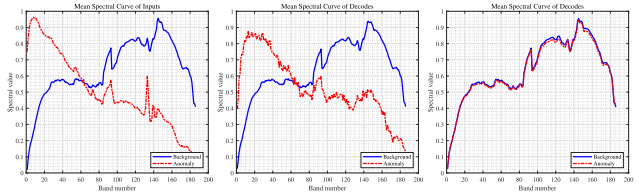
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(a) (b) (c)

Fig. 2. Comparison of the spectral curves obtaining from CAE and our proposed OTCMA. The blue solid lines and red dashed lines represent the average spectral curves of background and anomaly. (a) Input spectral curves. (b) Reconstructed spectral curves obtained by CAE (our model without memory module). (c) Reconstructed spectral curves obtained by OTCMA.

to reconstructing other class samples, such as anomalies [8]. It is especially the case for complex backgrounds. Because when an AE is trained on some of the backgrounds with complex shapes and structures, the latent space inherently learns to represent certain anomalies that may share some of the same features of the background. Second, due to low spatial resolution in HSI, mixing pixels inevitably contain some anomalous information, which helps to reconstruct the anomalies.

This article proposes a cluster-memory augmented deep AE via optimal transportation (OTCMA) for hyperspectral AD. To be specific, we solve the abovementioned problems from two aspects. To obtain more compacted features for each category, we enhance the feature consistency of the same category's samples and feature discrimination of different categories' samples. Consistency denotes that the features of samples in the same class are close, and discrimination means the features in different classes are distinct. Considering labels are unavailable in the AD task, we propose to leverage optimal transportation (OT)-based deep clustering [9] to learn the consistency and discrimination features in an unsupervised manner. By adding an extra cluster layer, we introduce the OT to estimate pseudo-labels [10]. Then, these pseudo-labels are utilized to learn the consistent and discriminative features by minimizing a cross-entropy loss.

To prevent the interference of the anomalous mixed in the training data and further restrict over reconstruction ability of the AE, we propose a memory module. The memory module comprises a memory bank and reading and writing operations. The memory bank is utilized to store the memory prototypes of different samples. The reading and writing operations are, respectively, responsible for retrieving content from the memory bank and updating the memory prototypes of the memory bank. In this work, we define that the memory prototypes are the cluster centers. As shown in the lower part of Fig. 1, a spectral pixel no longer depends on its own encoded features for reconstruction but reads the corresponding memory prototypes in the memory bank for reconstruction. Because anomalies account for only a tiny part of an HSI, building on cluster centers, the memory prototypes tend to represent different kinds of backgrounds and are the most consistent features for each category background. Therefore, indirect sample reconstruction with the help of a memory module can more effectively control the reconstruction ability of the AE and prevent anomalies from being reconstructed.

By enhancing the discrimination and consistency of learned features by deep clustering based on OT and further controlling the reconstruction ability by the memory model, the AE is restricted only well to reconstructing background, which more closely follows the assumptions of the AE-based detector. As shown in Fig. 2(a) and (c), depending on the memory module, the background can be well reconstructed, while the anomaly is mistakenly reconstructed into the background, resulting in a large reconstruction error. The main contributions of this article are concluded as follows.

1) We propose to learn more compacted features to prevent features of one category from contributing to reconstructing other categories by enhancing feature consistency and discrimination. By OT-based deep clustering, similar samples are closed in the latent space while different samples are kept apart.
2) We propose a memory module to prevent the interference of the anomalous mixed in the training data and further restrict over reconstruction ability of the AE. By retrieving corresponding memory prototypes (cluster centers) from the memory bank, samples are not directly reconstructed by their own encoded features. This strategy further restricts the introduction of more information into network training, avoiding reconstructing anomalies.

The remainder of this article is organized as follows. The related work about HSI-AD tasks is presented in Section II. Our proposed OTCMA model is detailed in Section III. The experimental results and the discussion are given in Section IV. We finally conclude our work in Section V.

II. Related Work

A. AD in HSI

The AD method in HSI is to separate the spectral pixels that stand out from the cluttered background [2], [11]. In practice, there are usually many different land covers in a scene, and there is no prior knowledge of spectral signature about both anomaly and background. Therefore, these methods are intrinsically unsupervised, which deal with the cases in multiclass backgrounds contaminated by anomaly [12], [13]. In recent years, more and more AD methods in HSI have been proposed. For clarity, we divided them as traditional methods and deep learning-based methods. The former commonly utilizes statistical representation and dictionary learning to model the background, and the latter builds the model or learning features of the background by the deep neural network.

B. Traditional Methods

As early as the 1990s, [14] and [15] investigated a statistical anomaly detector called the Reed-Xiaoli (RX) detector to detect the anomaly and assumed that the background of an HSI follows the zero-mean multivariate normal distribution. The detector is derived in the form of Mahalanobis distance,

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which measure the difference between testing pixels and local or global background. Later, various modified versions of the RX detector were proposed to adapt the cases of non-Gaussian distributions and high-dimension data, such as linear RX [16], kernel RX [17], and regularized RX [18].

Support vector data descriptor (SVDD) [19], [20] is another representative method that avoids density estimation and precise distribution modeling. By exploiting the "kernel trick," it can accurately model the support of nontrivial multimodal distribution and yield powerful detection ability in the cases of non-Gaussian and non-homogeneous backgrounds.

For another branch, the representation and dictionary learning-based detectors are widely investigated. Leveraging collaborative representation (CRD) [21] proposed the detector by assuming that the central pixels can be well represented by their neighbor while the anomaly cannot. According to the truth, the background in a scene is mostly homogeneous, which shows the low-rank property, and the anomaly is the minority, which shows sparsity, respectively. Therefore, low-rank and sparse-based methods, including low-rank and sparse matrix decomposition-based Mahalanobis distance detector (LRaSMD) [22], low-rank and sparse representation based detector (LRSR) [23], potential anomaly and background dicrionaries construction based detector (PAB-DC) [24], spectral unmixing and dictionary-based low-rank decomposition detector (ADLR) [25], [26], and spectral difference low-rank representation learning based detector (SDL-RDL) [27], spring up and yield more powerful detection performance. Although these methods are non-parametric and free from modeling the data distribution, the dictionary and representations must be learned globally. All samples need to join into the costly SVD decomposition process, which leads to the high requirement of the devices. In addition, the learned model is difficult to generalize to other datasets, and each HSI dataset needs to compute the SVD decomposition independently.

C. Deep Learning-Based Methods

With the development of deep learning, there are many deep learning methods for various remote sensing image processing tasks, such as image registration [28], change detection [29], [30], image classification [31]–[34], image segmentation [35], and object detection [36]–[38]. In the early phase of deep learning-based HSI processing revival, [39] proposed to transform the AD task into a classification problem. It first builds the positive (pixels from the same classes) and negative pixel pairs (pixels from different classes) and then uses the convolution neural network (CNN) to learn a deep classification model to recognize the positive and negative pixel pairs [40]. In general, the background and its neighbor pixels are easily classified into positive pixel pairs, while the anomaly and its neighbor pixels tend to be classified into negative pixel pairs. After that, a series of milestone works based on AE was proposed in [41]–[43] and [44]. In spectral constraint adversarial autoencoders (SC-AAE) [41], an adversarial AE is trained to learn discriminating features by minimizing both the reconstruction error and spectral angle mapper (SAM). Data preprocessing and bilayer architecture with morphological operations enhance the discrimination between background and anomaly. By introducing fully CAE with skip connection, [44] proposed a specifically designed framework Auto-AD to direct learn spatial–spectral characteristics. An adaptive-weighted strategy is proposed to suppress the anomaly reconstruction further to filter potential anomaly pixels in the training process. In the end, the anomaly is directly detected by the reconstruction error.

A generative adversarial network (GAN) was introduced to HSI-AD task in [45], where an AE is equipped with two discriminators; one discriminator is utilized to push the learned feature into the embedding space following multivariate Gaussian distribution to learn consistent features of the background, and the other one is utilized for pixels reconstruction. The same idea is also utilized in [46].

It should be noted that there are many undetected anomaly pixels according to the reconstruction error. The main reason is that the anomaly can be well reconstructed even if they are the minority, which is also proved in [45] and [46]. Different from [45] who push the learned features into a Gaussian space, we learn feature consistency for each kind of land cover using a memory model. Compared with the AE-based method, we introduce OT-based deep clustering to learn discriminating features by pulling intra-class samples together and pushing inter-class samples apart, which is the foundation of the success of the memory module.

III. Proposed Framework

The framework of our proposed cluster-memory augmented deep autoencoder via optimal transportation (OTCMA) is shown in Fig. 3. As the backbone of our model, CAE is trained to reconstruct the pixels. Deep clustering based on OT is proposed for feature consistency and discrimination learning. The inputs of the memory module are the cluster predictions and the encoded features of the encoder. The memory module is responsible for indirectly reconstructing samples preventing the interference of the anomalies and further restricting the reconstruction ability of the AE. In the following part, we will first present the formulation of HSI-AD task. We then detail the main framework of OTCMA, including the backbone of our model (CAE), OT-based deep clustering, and cluster-based memory augmented AE, separately.

A. Problem Formulation

We consider an HSI $I \in \mathbb{R}^{h\times w\times d}$, where d is the number of the spectral bands, and h and w are the height and the width of the data, respectively. The 2-D spectral pixels set is formulated as $X^{N\times d} = \{x_i\}$, where $N = w \times h$. It contains two disjoint set X_{out} and X_{in}. X_{out} and X_{in} represent the set of anomaly and the background, and they satisfy $X = X_{\text{in}} \cup X_{\text{out}}$ and $X_{\text{in}} \cap X_{\text{out}} = \emptyset$. In our work, we try to learn features, which are consistent for intra-class pixels and discriminating for inter-class pixels, so that OTCMA can recognize X_{out} from X_{in}.

B. Convolutional AE

The CAE is composed of an encoder and a decoder. The encoder $f(\cdot)$ learns a mapping from the input x_i to feature $z_i \in \mathbb{R}^{d_z}$ in the embedding space, and the decoder $g(\cdot)$ aims

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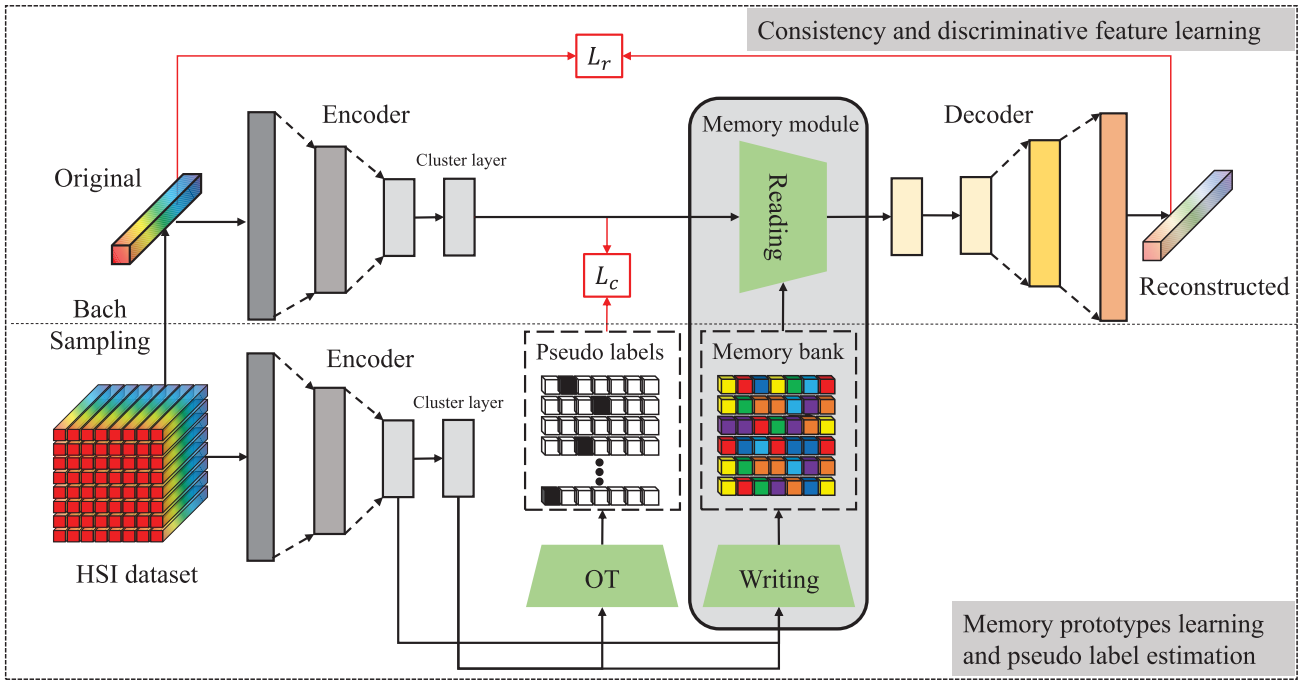


Fig. 3. Schematic illustration of our proposed OTCMA. It consists of one encoder with an additional cluster layer, one decoder, and a memory module. (Top) AE learning procedure. (Bottom) Pseudo-label updating process and memory updating process using OT and writing operation. In training, the pseudo-labels and memory prototypes update every few epochs utilizing the whole dataset. The inputs of the decoder are the memory prototypes reading from the memory bank with respect to the predictions of the cluster layer. The whole AE is learned using a reconstruction loss and a cross entropy loss whose inputs are original-reconstructed spectral pairs and predictions-pseudo-labels pairs, respectively.

to recover $\hat{x}_i$ from z_i. The feature learned by the encoder and reconstruction learned by the decoder can be defined as

$$\begin{aligned} z_i &= f(x_i) \\ \hat{x}_i &= g(z_i). \end{aligned} \tag{1}$$

The CAE is optimized by minimizing the reconstruction error between the input and the output of the decoder, which is formulated as

$$\mathcal{L}_e = \|g(z_i) - x_i\|_2^2. \tag{2}$$

AE-based AD methods try to recognize the anomaly according to the reconstruction errors. However, as shown in Fig. 2(b), it cannot achieve the desired results only by minimizing (2). We attribute its lack of success to restricting the over-reconstruction ability to anomaly pixels. In the following part, we try to build our model to learn more compacted features, e.g., more consistent and discriminative features, for each sample. A sample is no longer reconstructed by its own feature z_i, but its corresponding consistent features $\hat{z}_i$ retrieving from the memory module.

C. Deep Clustering by OT for Consistent and Discriminative Feature Learning

In our work, enhancing the feature consistency of the same categories' samples and the feature discrimination of the different categories' samples is key to preventing AE reconstructing anomalies. Because no labels are available in hyperspectral AD, deep clustering is utilized for unsupervised feature learning. We add another fully connected layer on the top of the encoder, which outputs the cluster predictions. In this work, we propose to utilize OT [9] to first estimate the pseudo-labels, which are then utilized for training the model as a common classification task.

Considering each pixels $x_i \in X^{N\times d}$, it will be assigned a label y_i, which reflects the cluster assignment of pixel x_i. In our framework, we share the feature extraction part of the encoder $f(\cdot)$, so the cluster predictions are defined as

$$p(y = y_i|x_i) = \psi(h \circ f(x_i)) \tag{3}$$

where $y_i \in \{y_i\}^K$ denotes K clusters, ψ is the softmax function, and h is the clustering layer. If the labels are known, h and f are learned to obtain discriminating features by minimizing the average cross-entropy loss

$$\mathcal{L} = E(p|y_1, \ldots, y_K) = -\frac{1}{N}\sum_{i=1}^{N} \log p(y_i|x_i). \tag{4}$$

In our work, labels y_i in (4) are unknown. To solve this problem, y_i is replaced by the pseudo-labels, which is defined by the posterior distributions $q(y|x_i)$

$$\mathcal{L}_c = E(p|q) = -\frac{1}{N}\sum_{i=1}^{N} q(y|x_i) \log p(y_i|x_i). \tag{5}$$

Unfortunately, (5) cannot be directly optimized, because it will lead to degenerate results in which all pixels are grouped into one cluster, while this equation can be formalized to OT problem, which is a classical problem to minimize the "transport cost" between "suppliers" and "demanders" [10].

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In the classical OT problem [47], [48], there are two roles: suppliers and demanders. Different demanders can put forward supply–demand to the supplier, and the supplier needs to supply goods from different demanders. However, when the supplier provides goods for each demander, it needs to bear the corresponding transportation cost. Therefore, the supplier needs to design and plan reasonable goods distribution methods according to the requirements of each demander to minimize the transportation cost while meeting the needs of all demanders. In the process of pseudo-label estimation, for each sample, converting the clustering layer's outputs into pseudo-labels is similar to transporting goods from the supplier to the demander in OT.

In our model, the number of suppliers and the demanders is equivalent to the output number of the cluster layer (the cluster number), and the number of goods is equivalent to the number of samples in HSI. One needs to transport all goods from the suppliers to each demander satisfying, minimizing the transportation cost, in other words, grouping all samples into K clusters with minimum transportation cost. By assuming the data can be equally grouped into several clusters, the posterior distributions $q(y|x_i)$ are relaxed to be an element of the transportation polytope, which satisfies

$$U(u, v) := \left\{ Q \in R_+^{N\times K} \middle| Q\mathbf{1} = u,\, Q^\top \mathbf{1} = v \right\} \tag{6}$$

where Q is an $N \times K$ transport matrix, and each element $Q_{i,y} = (1/N)q(y|x_i)$ represents the ith cluster posterior probability prediction with respect to cluster y. u and v are the marginal distribution that is defined to uniformly split the pixels

$$u = \frac{1}{N} \cdot \mathbf{1}, \quad v = \frac{1}{K} \cdot \mathbf{1} \tag{7}$$

where $\mathbf{1}$ are the vectors of all ones with the appropriate dimensions.

In this way, (5) can be written as a OT problem with transport matrix Q and cost matrix $-\log P$.

$$\min_{Q\in(u,v)} \langle Q, -\log P\rangle + \frac{1}{\lambda}\mathrm{KL}(Q||uv^\top) \tag{8}$$

where P is the $N \times K$ prediction matrix obtaining from (3), whose elements are defined as $P_{i,y} = (1/N)p(y|x_i)$.

In the OT problem, the negative logarithm form of P is defined as the "transport cost." The total transportation cost in (8) is minimized by finding an OT matrix Q, which is the optimal pseudo-label in our framework. The Kullback–Leibler (KL) divergence regularization term is added, so that the model can be solved fast by the Sinkorn–Knopp algorithm [10]. Following this way, the optimal pseudo-label Q^* is

$$Q^* = \mathrm{diag}(\alpha)P^\lambda \mathrm{diag}(\beta) \tag{9}$$

where α and β are scaling coefficients preserving Q as a probability matrix. In each iteration, they are updated by

$$\begin{aligned} \forall y : \alpha_y &\leftarrow [P^\lambda \beta]_y^{-1} \\ \forall i : \beta_i &\leftarrow [\alpha^\top P^\lambda]_i^{-1} \end{aligned} \tag{10}$$

where λ is the parameters to control the convergence speed of the OT process, and it is empirically set as 10 in all our experiments by default.

In the training process, (5) and (8) are optimized alternatively for model updating and pseudo-label estimation. The procedure is detailed in Algorithm 1. The obtained features and cluster prediction are the input of subsequent memory module learning.

D. Memory Module

To prevent the interference of the anomalous mixed in the training data and further restrict over reconstruction ability of the AE, we propose a memory module. As shown in Fig. 3, the memory module is composed of a memory bank and reading and writing operations. The memory bank stores the prototypes, which are the dynamically updatable centers for each cluster. The writing and reading operations are responsible for updating the memory bank prototypes and retrieving them for sample reconstruction, respectively.

1) Reading Operation: As shown in Fig. 3. The reading operation aims to take out corresponding prototypes of current pixels stored in the memory bank $M \in \mathbb{R}^{d_z\times K}$, where the jth column $m_j \in M$ is the prototype of cluster j, and d_z is the dimension of the prototype. The operation needs to be derivable, so that the loss from the memory module can be backpropagated to the cluster layer $h(\cdot)$ and the encoder $f(\cdot)$. Given the cluster predictions p_i, the reading operation $r(\cdot)$ is formulated as an attention-like operation to obtain $\hat{z}_i$ in a soft way

$$\hat{z}_i = r(p_i, M) = M \cdot p_i. \tag{11}$$

Instead of directly utilizing itself feature z in (1) to reconstruct the input pixel, we take the learned feature $\hat{z}_i$ from the memory module to recover the input pixel. Also, the reconstruction loss is rewritten as

$$\mathcal{L}_r = \|g(\hat{z}_i) - x_i\|_2^2. \tag{12}$$

2) Writing Operation: As shown in Fig. 3, the writing operation aims to update the prototypes stored in the memory bank dynamically. After obtaining the predictions p_i and features z_i of the whole scene, the prototypes are also calculated in a soft way with the writing operation $w(\cdot)$ formulating as

$$M = w(Z, P) = Z^\top \cdot P \tag{13}$$

where $Z \in \mathbb{R}^{N\times d_z}$ is the features of all samples.

Intuitively, the prototype of a cluster is calculated by the weighted summation of all samples with respect to corresponding predictions belonging to the current cluster.

E. Train and Inference

In the training process, to balance the detection performance and optimization efficiency, the optimization process of the pseudo-label estimation using OT and the memory prototype update according to the current clustering results will be calculated once every several epochs. Therefore, the overall optimization process of the model can be divided into two stages. All samples are used for pseudo-label estimation, and memory prototypes update in the first stage. The pseudo-labels are estimated by (8), and the memory prototypes are updated

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Algorithm 1 Training Algorithm of OTCMA

Input: Training samples $X \in \mathcal{X}$, the number of memory prototypes K,
Output: A trained MCOD with parameters sets Θ
for *each epoch* **do**
  if *its time to update M and y* **then**
    Step1: Calculate pseudo labels for all samples by the OT using Eq. (8);
    Step2: Write the memory prototypes into memory bank by Eq.(13) using current cluster predictions for all samples.
  end
  for *each iteration* **do**
    Step1: Forward a random mini-batch of samples to obtain feature z and cluster prediction p;
    Step2: Conduct reading operation $r(\cdot)$ based on current memory prototypes M using Eq. (11) to get $\hat{z}$;
    Step3: Learn to reconstruction samples by minimizing $\mathcal{L}_r$ in Eq. (12);
    Step4: Given current pseudo labels, learn discriminative and consistent features and cluster prediction by minimize $\mathcal{L}_c$ in Eq. (5).
  end
end

by (13). The second stage is the network learning procedure. For each iteration, features and the AE are updated together by (5) and (12). The total loss is

$$\mathcal{L} = \mathcal{L}_c + \mathcal{L}_r. \tag{14}$$

The training process of OTCMA is summarized in Algorithm 1. The model is trained in a transductive way for AD, which means that the unlabeled training data are also the test data. The model is trained by utilizing all samples of the HSIs (including background pixels and anomalies), and then, the trained model is used to detect the anomalies in this image.

For inference, the commonly used anomaly score in the AE model is reconstruction error. It is the mean-square error between the original inputs and the reconstructed spectral pixels

$$s_1 = \|g(\hat{z}_i) - x_i\|_2. \tag{15}$$

However, the reconstruction error only considers single-pixels' reconstruction performance and lacks stability and robustness. Conversely, Mahalanobis distance measures the difference between the distribution of training data and the test spectral pixels in the feature space. Given the encoded features z of a test pixel x, the Mahalanobis distance can be formulated as

$$s_2 = (z - \mu)^\top \Gamma^{-1} (z - \mu) \tag{16}$$

where μ and Γ are mean vector and covariance of all z.

This work finds that the anomaly score obtained from Mahalanobis distance is more robust and effective on relatively bigger-size anomalous objects and complex scenes, while the reconstruction error shows better performance on small and isolated objects. Therefore, we take the combination of Mahalanobis distance and reconstruction error as the final anomaly score

$$s = \eta s_1 + (1 - \eta) s_2 \tag{17}$$

where η is a balance parameter, which is set as 0.6 in our experiment; more analyses are given in Section IV.

TABLE I
DETAILED DESCRIPTION OF THE ABU DATASET

Data name	Captured place	Resolution	Size
Airport-1	Los Angeles	7.1m	$100 \times 100 \times 205$
Airport-2	Los Angeles	7.1m	$100 \times 100 \times 205$
Airport-3	Los Angeles	7.1m	$100 \times 100 \times 205$
Airport-4	Gulfport	3.4m	$100 \times 100 \times 191$
Beach-1	Cat Island	17.2m	$150 \times 150 \times 188$
Beach-2	San Diego	7.5m	$100 \times 100 \times 193$
Beach-3	Bay Champagne	4.4m	$100 \times 100 \times 188$
Beach-4	Pavia	1.3m	$150 \times 150 \times 102$
Urban-1	Texas Coast	17.2m	$100 \times 100 \times 204$
Urban-2	Texas Coast	17.2m	$100 \times 100 \times 207$
Urban-3	Gainesville	3.5m	$100 \times 100 \times 191$
Urban-4	Los Angeles	7.1m	$100 \times 100 \times 205$
Urban-5	Los Angeles	7.1m	$100 \times 100 \times 205$

IV. EXPERIMENT

In the following part, we first demonstrate the superiority of our OTCMA. We compare our method with four baselines and four state-of-the-art methods, and we then analyze the performance of different anomaly measurements. To further explore the effectiveness of our memory module, we conduct HSI-AD based on different training settings: with and without anomalous pixels in the training dataset. More detailed and intermediate results are presented to show more insight of our model. The benchmark dataset used in our experiments are: Airport–Beach–Urban (ABU) dataset [49], San Diego dataset, and hyperspectral digital imagery collection experiment (HYDICE) dataset [50]. All the experiments are conducted on the workstation with Intel Xeon Processor E5-2630 v3 2.40 GHz $\times$ 20 and NVIDIA GTX1080.

A. Dataset

1) ABU Dataset: It contains 13 HSIs of the scene airport, beach, and urban, which was built in [49]. The scenes were captured by the airborne visible/infrared imaging spectrometer (AVIRIS) sensor and the reflective optics system imaging spectrometer (ROSIS-03) sensor. The size of the images is ranging from 100 $\times$ 100 to 150 $\times$ 150. We present the detailed information of these 13 images in Table I. The false-color images and the ground truth are shown in Figs. 4, 5, and 6(a) and (b).

[1]For Airport-1, 2, 3, 4; Beach-1, 2, 3, 4; Urban-1, 2, 3, 4, 5; San Diego-I, II, and HYDICE * is 25, 25, 25, 23; 23, 24, 23, 12; 25, 25, 23, 25, 25; 23, 23; 21.

[2]The corresponding ** for each images is 6, 6, 6, 8; 8, 2, 5, 7; 5, 8, 8, 6, 6; 8, 8; 8.

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TABLE II
NETWORK STRUCTURE OF OTCMA

Model name	Layer name	Form
Encoder	Block1 64	[Conv1D(1×9) BN ReLU]$\times 2$
	Block2 128	[Conv1D(1×9) BN ReLU]$\times 2$
	Block3 256	[Conv1D(1×9) BN ReLU]$\times 2$
	Block4 512	[ConV1D(1×9) ReLU]$\times 2$
	Fc[1]	Linear $(* \times 512) \times 1024$ ReLU
Cluster layer	Fc	Linear 1024×10
Decoder	Fc[1]	Linear $1024 \times (* \times 512)$
	Block5 256	[Conv1D(1×9) BN ReLU]$\times 2$
		[ConvT1D(1×9) BN ReLU]
	Block6 128	[Conv1D(1×9) BN ReLU]$\times 2$
		[ConvT1D(1×9) BN ReLU]
	Block7 64	[Conv1D(1×9) BN ReLU]$\times 2$
		[ConvT1D(1×9) BN ReLU]
	Block8 1^2	[ConvT1D ($1 \times **$) BN ReLU]
		[Conv1D(1×9) ReLU]
		[Conv1D(1×9) ReLU]

2) San Diego Dataset: The dataset is provided in [23], which was collected by AVIRIS over San Diego, CA, USA. It contains two 100×100 HSI images with 224 spectral bands ranging from 370 to 2510 nm. The anomalous spectral pixels are the body and edges of three aircraft. To use this dataset, we first remove the bands (1–6, 33–35, 97, 107–113, 153–166, and 221–224) contaminated by noise or useless due to water absorption. In our experiments, these two images are marked as San Diego-I and San Diego-II.

3) HYDICE Dataset: The dataset is obtained by the HYDICE sensor. The size of this image is 80×100 with a spatial resolution of 1 m. It contains 175 spectral bands of spectral resolution 10 nm. The 21 anomalous pixels are the cars and roofs [50].

B. Implementation Details

In the experiment, we propose a spectral CAE building by 1-D convolution and fully connected layers. The network is built by one encoder, one decoder, and one cluster layer. The encoder and decoder contain four convolution blocks and one fully connected layer. The cluster layer is a fully connected layer that inputs feature outputting from the encoder to produce cluster predictions. The structure is given in Table II. It should be noted that the input size of the *fc-layer* (*) in encoder and decoder and the transpose convolutional layer of *block-8* (**) are varied for different datasets. Besides these parameters, the size of all other layers is fixed to be the same for different datasets. The output size of the encoder is 1024, and we set the number of prototypes as 5 for all datasets. The model is optimized by Adam optimizer with $\beta = (0.9, 0.999)$ and weight decay $= 0.0005$. We set the base lr $= 0.001$ with a cosine decay schedule for 200 epochs. Following the common setting in hyperspectral AD, all methods utilize HSIs mixed with anomalies for both training and testing.

To evaluate the detection performance, we adopt the receiver operating characteristic (ROC) curve [51], and we report the numerical results of the area under the ROC curve [Area under receiver operating characteristic curve (AUROC)]. The detection map is visualized with colors ranging from dark blue to bright red. The objects with higher anomaly scores will be brighter on the detection map.

C. AD Performance Comparisons on ABU Dataset

In this section, we compare our OTCMA with four baselines: Global-RX [15], Local-RX [15], SVDD [19], and CAE on the ABU dataset. For Global-RX, Local-RX, and SVDD, we follow the same parameter settings reported in [21] and [43]. To show the effectiveness of our proposed memory module, we add the comparisons with CAE, which has a similar setting to our framework but without the memory module. Moreover, we compare our method with collaborative representation-based method CRD [21] and morphology-based method attribute and edge-preserving filters based detector (AED) [49], which are the state-of-the-art traditional methods. The comparison with the state-of-the-art deep AD method hyperspectral image anomaly detection via generative adversarial network (HADGAN) [45], autoencoding adversarial network (AEAN:) [52], and detector based on Skewed t-distribution and autoencoder (MVSkt) [53] is also presented. In particular, HADGAN [45], AEAN [52], and MVSkt [53] build upon adversarial AE. In HADGAN [45], it also constrains the feature consistency in the latent space by mapping both anomaly and background pixels into the same position. AEAN [52] ensembles 1-D, 2-D, and 3-D AE for spectral–spatial feature learning and finally detects anomalies based on weighted RX. In MVSkt [53], multivariate student's (MVSt) distribution is used for background modeling. For HADGAN, AEAN, and MVSkt, we follow the experiments settings reported in [45], [52], and [53].

For clear and fairly comparison, we report the detection results only using spectral information: OTCMA-R, OTCMA-M, and OTCMA-Spe. OTCMA-R and OTCMA-M are the detection results using reconstruction error and Mahalanobis distance. OTCMA-Spe is calculated by (17), which is the ensemble of these two scores. We also present the results of integrating the spatial feature into our OTCMA, which is marked as OTCMA-E. To use the spatial feature, we add the postprocessing based on morphological attribute filter [43] to our model. The numerical results about the AUROC values are shown in Table III, and the detection map is given in Figs. 4–6, respectively.

1) Comparisons With Baselines: In terms of the results obtained by Global-RX, Local-RX, SVDD, and OTCMA-Spe in Table III, our method achieves better performance. According to the average results of each scene, our method OTCMA-Spe surpasses Global-RX, Local-RX, and SVDD by 8.61%, 3.36%, and 1.39% on the scene of airport, 2.8%, 1.13%, and 2.02% on the scene of beach, and 0.76%, 2.18%, and 2.47% on the scene of urban, respectively. In terms of total-mean results, OTCMA-Spe that only uses spectral information can improve the HSI-AD results by 3.8%, 2.22%, and 2.11% compared with Global-RX, Local-RX, and SVDD, respectively. These results demonstrate that OTCMA shows significant superiority over these baselines. The performance gains come from two respects. On the one hand, we do not need to precisely model the data distribution, which is always impractical. On the other hand, our method learns more informative features to detect outliers.

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TABLE III
COMPARISON WITH SEVEN METHODS ON ABU DATASET. THE BEST AND THE SECOND-BEST PERFORMANCE OF EACH DATASET IS SHOWN IN BOLD FONT AND UNDERLINED

Dataset	Global-RX	Local-RX	SVDD	CRD	AED	CAE	HADGAN	AEAN	MVSkt	OTCMA-R	OTCMA-M	OTCMA-Spe	OTCMA-E
Airport-1	82.21	94.58	95.01	95.77	**99.23**	80.45	98.50	97.63	95.56	74.32	95.66	95.53	98.20
Airport-2	84.04	98.74	99.00	97.44	99.36	90.21	**99.59**	99.59	99.47	92.07	96.21	97.82	99.12
Airport-3	92.88	94.67	90.80	95.64	97.56	80.94	98.43	98.32	92.07	92.10	94.31	96.56	**98.84**
Airport-4	95.26	87.40	98.48	97.57	99.53	97.16	98.83	99.43	98.59	98.86	98.47	98.93	**99.60**
Airport-Mean	88.60	93.85	95.82	96.61	98.92	87.19	98.84	98.74	96.42	89.34	96.16	97.21	**98.94**
Beach-1	98.07	99.56	95.83	99.16	99.74	98.52	**99.95**	99.81	99.55	97.62	99.36	99.42	99.61
Beach-2	91.06	97.77	96.20	94.16	98.25	92.42	98.77	**98.89**	90.41	98.04	92.56	98.19	98.63
Beach-3	99.99	99.97	99.99	99.96	99.97	98.26	99.99	99.99	99.97	99.28	99.90	99.96	**99.99**
Beach-4	95.38	93.91	95.61	94.50	99.73	82.59	**99.92**	99.36	99.52	96.10	98.25	98.15	99.90
Beach-Mean	96.13	97.80	96.91	96.95	99.42	92.94	**99.66**	99.51	97.36	97.76	97.50	98.93	99.53
Urabn-1	99.07	99.68	98.62	99.48	99.80	97.45	99.71	99.79	99.48	99.54	96.84	99.38	**99.83**
Urban-2	99.46	92.31	93.56	94.10	99.85	97.06	99.70	99.87	98.36	98.34	97.45	99.64	**99.92**
Urban-3	95.13	98.00	96.58	96.34	99.76	97.86	99.37	**99.86**	96.58	98.06	93.93	97.40	99.55
Urban-4	98.87	96.96	96.21	98.16	99.12	88.23	99.54	99.65	93.14	98.73	84.58	98.92	**99.74**
Urban-5	96.92	95.38	94.49	95.21	98.45	76.23	**99.43**	98.72	96.20	96.20	95.28	97.94	99.19
Urban-Mean	97.89	96.47	95.89	96.66	99.40	91.37	99.55	99.57	96.75	98.17	93.62	98.65	**99.65**
Total Mean	94.49	96.07	96.18	96.73	99.26	90.57	99.36	99.30	96.83	95.33	95.60	98.29	**99.39**

As shown in Table III, OTCMA-Spe yields about 10.02%, 5.99%, and 7.28% improvements on the scene of airport, beach, and urban than CAE. Moreover, OTCMA-Spe obtains 7.71% performance gains on average. Both OTCMA-R and OTCMA-M achieve better performances compared with CAE. These results demonstrate that OTCMA can learn more consistent and discriminative features in the training process, which effectively mitigates the negative effects of anomalous mixed in the training data.

2) Comparisons With State of the Arts: This part compares our proposed method with state of the art on HSI-AD, and the results are listed in Table III. As discussed in [43] and [45], spatial information can further boost the performance. So, we add a morphological attribute spatial feature learning as a postprocessing operation following the settings in [43]. First, we can see that our proposed method based on cluster memory acquires the best performance. OTCMA-E achieves better average performance than CRD, AED, HADGAN, AEAN, and MVSkt. Especially, OTCMA-E surpasses CRD by 2.33%, 2.58%, and 2.99%, and OTCMA-Spe surpasses CRD by 0.6%, 1.98%, and 1.99% on the scene of airport, beach, and urban. The gap in the total results between OTCMA-Spe and CRD is 1.56%, enlarging to 2.66% when adding spatial characteristics in OTCMA-E.

Although AED only relies on spatial characteristics, it shows surprising detection performance. So, it is important to introduce the spatial characteristics into the training process for the HSI-AD task. When we add the postprocessing based on the morphological attribute filter [43], the mean result of our OTCMA increases from 98.29% (OTCMA-Spe) to 99.39% (OTCMA-E). In particular, compared with OTCMA-Spe, OTCMA-E obtains 1.73%, 0.6%, and 1.0% performance gains on the scene of airport, beach, and urban, respectively. The detection maps in Figs. 4(f)–6(f) show that most of the backgrounds are suppressed, and the anomaly reflects a strong response compared with the background. The reason is that the morphological attribute features obtained by closing and opening operations can effectively preserve small objects in the scene. The same content in the opening and closing map will be removed in a differential map between them, which will effectively eliminate most of the background.

Compared with deep methods HADGAN, AEAN, and MVSkt, OTCMA-E obtains 0.03%, 0.09%, and 2.56% on average performance gains, respectively. By integrating the spatial postprocessing or learning ensemble AE based on 1-D, 2-D, and 3-D convolutional networks, HADGAN and AEAN can achieve extremely high detection accuracy. Our proposed OTCMA-E can be slightly better than HADGAN and AEAN. Even if only 1-D CAE is utilized for feature learning, OTCMA-Spe gets better performance than MVSkt in which 3-D CAE is used for spectral–spatial feature learning. These results further prove the effectiveness of our model.

D. Analysis of Different Anomaly Scores

In this section, we discuss the AD performance obtained from different anomaly scores, including reconstruction error (OTCMA-R), Mahalanobis distance (OTCMA-M), and their ensemble (OTCMA-E). Then, we analyze the optimal ensemble parameters.

1) Performance of Different Anomaly Scores: From the results of Table III and Fig. 7, we can see that the OTCMA-M achieves slightly better performance than OTCMA-R on average. According to the detection map shown in Figs. 4(e)–6(e), the Mahalanobis distance is more sensitive to relatively larger anomalous objects. The main reason is that the distribution of all training features can be used to measure the anomaly in Mahalanobis distance, which is more stable and robust than the reconstruction error of every single pixel. Another reason is that our model can learn discriminating features by deep clustering. To prove this, we present the clustering visualization map in Figs. 4(c)–6(c). It can be seen that our method can separate different land covers into different clusters, which demonstrates that the learned features are discriminating. The discriminative features are beneficial to separating anomalies from backgrounds in the feature space.

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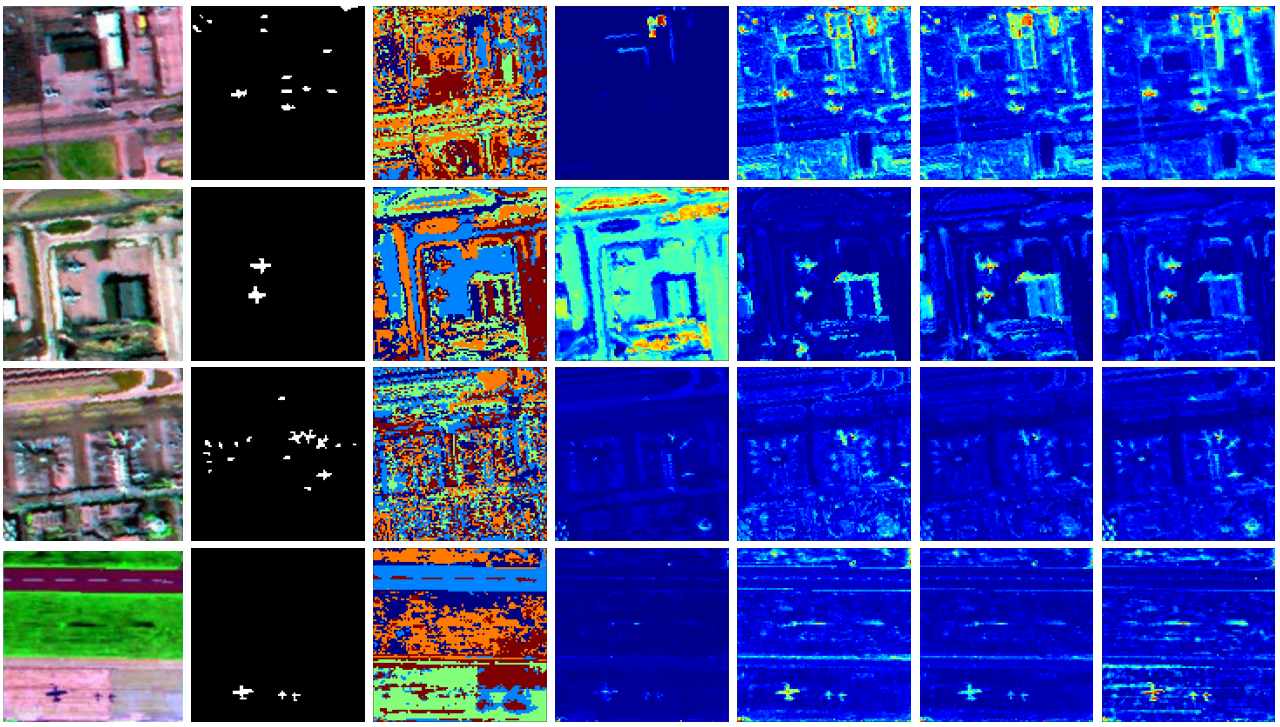


Fig. 4. Image descriptions and detection maps of Airport 1–4. (a) False color images. (b) Corresponding ground truth. (c) Cluster maps obtaining from the output of cluster layers. (d)–(g) Detection maps corresponding to OTCMA-R, OTCMA-M, OTCMA-Spe, and OTCMA-E, respectively.

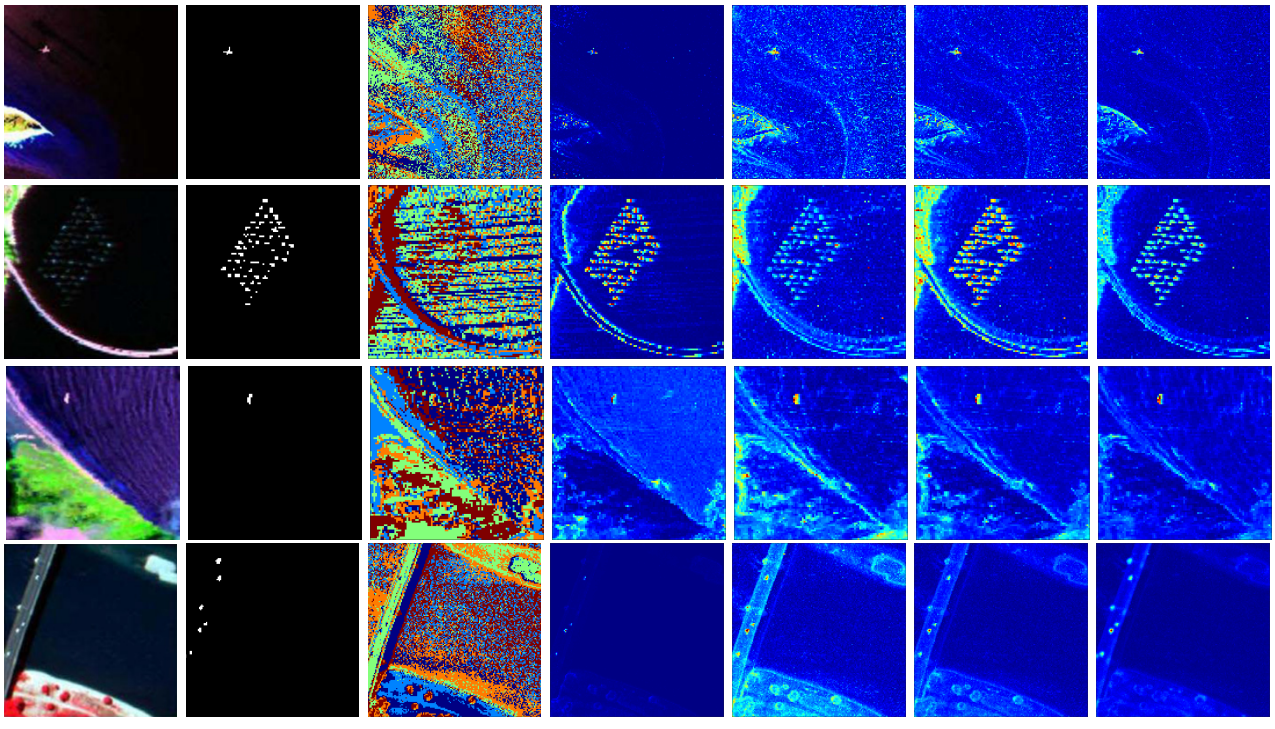


Fig. 5. Image descriptions and detection maps of Beach 1–4. (a) False color images. (b) Corresponding ground truth. (c) Cluster maps obtaining from the output of cluster layers. (d)–(g) Detection maps corresponding to OTCMA-R, OTCMA-M, OTCMA-Spe, and OTCMA-E, respectively.

From the results of Table III and Fig. 7, we can see that the OTCMA-M achieves slightly better performance than OTCMA-R on average. According to the detection map shown in Figs. 4(e)–6(e), the Mahalanobis distance is more sensitive

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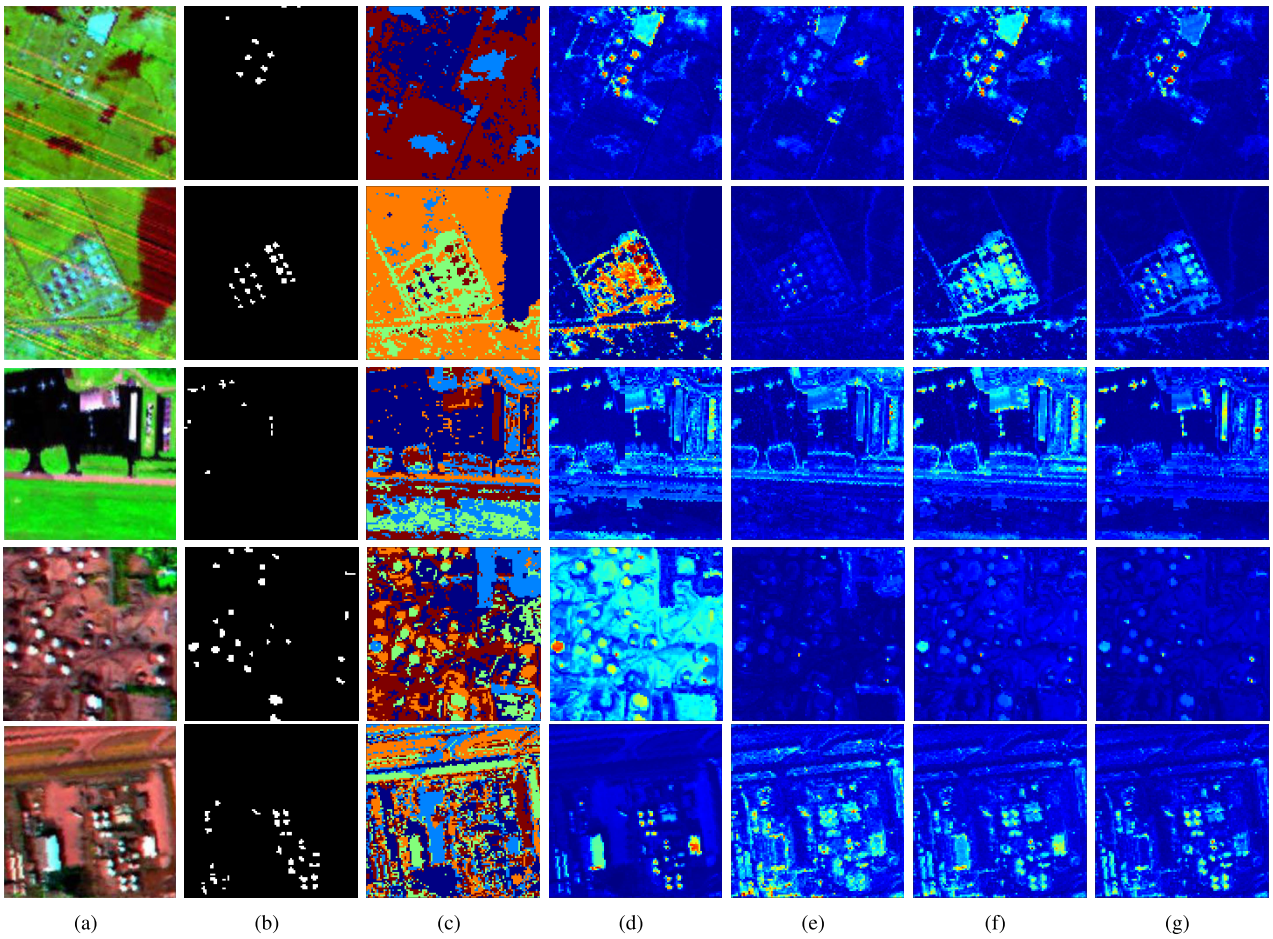


Fig. 6. Image descriptions and detection maps of Urban 1–5. (a) False color images. (b) Corresponding ground truth. (c) Cluster maps obtaining from the output of cluster layers. (d)–(g) Detection maps corresponding to OTCMA-R, OTCMA-M, OTCMA-Spe, and OTCMA-E, respectively.

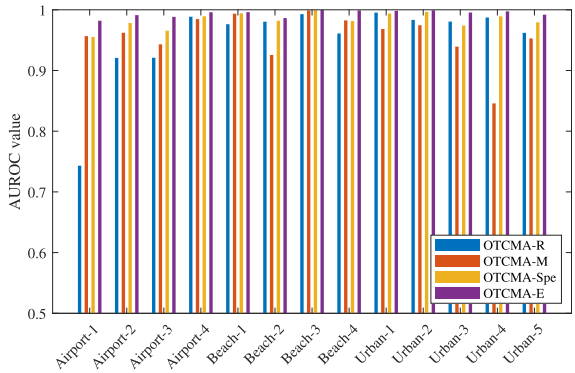


Fig. 7. Comparisons of OTCMA-R, OTCMA-M, OTCMA-Spe, and OTCMA-E.

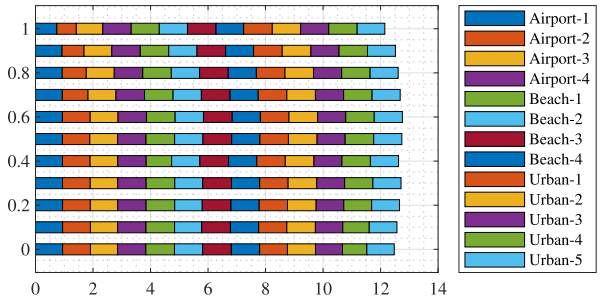


Fig. 8. AUROC values obtained by varying the ensemble parameters from 0 to 1.

to relatively larger anomalous objects. The main reason is that the distribution of all training features can be used to measure the anomaly in Mahalanobis distance, which is more stable and robust than the reconstruction error of every single pixel. Another reason is that our model can learn discriminating features by deep clustering. To prove this, we present the clustering visualization map in Figs. 4(c)–6(c). Our method can separate different land covers into different clusters, which demonstrates that the learned features are discriminating. The discriminative features are beneficial to separating anomalies from backgrounds in the feature space.

The results of Table III and Fig. 7 show that the combination of reconstruction error and Mahalanobis distance achieves better performance than their own performances. In particular, the average ROAUC results of OTCMA-Spe on the scenes of airport, beach, and urban are 97.21%, 98.93%, and 98.65%, respectively. OTCMA-Spe increases the ROAUC results of OTCMA-R by 7.87%, 1.17%, and 0.48% and improves the performance of OTCMA-M by 1.05%, 1.43%, and 5.03%. By comparing the detection map of OTCMA-Spe

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TABLE IV
COMPARISONS OF OTCMA AND CAE LEARNED WITH AND WITHOUT ANOMALY PIXELS. W DENOTES WITH ANOMALOUS PIXELS AND W/O DENOTES WITHOUT ANOMALOUS PIXELS IN THE TRAINING DATASET

Data name	CAE-W	CAE-W/O	OTCMA-W	OTCMA-W/O
San Diego-I	98.01	98.55	99.29	**99.55**
San Diego-II	96.83	96.86	98.65	**99.74**
HYDICE	99.70	99.81	99.74	**99.99**

[Figs. 4–6(f)] with OTCMA-R [Figs. 4–6(d)] and OTCMA-M [Figs. 4–6(e)] in Figs. 4–6, we can see that the anomalous objects show more strong response in OTCMA-Spe. Therefore, we utilize the combination of reconstruction error and Mahalanobis distance as our AD score.

2) Analysis of the Ensemble Parameters of η: To detect the anomaly objects with different sizes, we ensemble the reconstruction error and Mahalanobis distance as the anomaly score by 17. In this part, we analyze the HSI-AD performance varied the ensemble parameters η from 0 to 1 and conduct the experiment on the ABU dataset. The results are presented in the form of a stacked bar graph in Fig. 8. When $\eta = 0$ and $\eta = 1$, the merging anomaly score degenerates to a single Mahalanobis distance and a single reconstruction error. According to the mean results of all images in the ABU dataset, Mahalanobis distance and reconstruction error ($\eta = 0$ and $\eta = 1$) acquire worse performance than the ensemble anomaly score, and when $\eta = 0.6$, the performance reaches the peak. Therefore, the ensemble parameter is set as 0.6 in all experiments.

E. Analysis of Memory Module

In this section, we first analyze the effectiveness of the memory module under different training settings, with and without anomalous pixels in the training dataset. Then, we present a more careful analysis by visualizing the reconstructed spectral curves, the t-Distributed Stochastic Neighbor Embedding (T-SNE) results of the learned features, and the reconstructed results of memory prototypes to show more insightful studies about our proposed model in HSI-AD task. Finally, we analyze the HSI-AD results based on the varied number of prototypes.

1) Performance With and Without Anomaly in the Training Dataset: To test the effectiveness of the memory module, we compare our proposal with CAE based on different training settings: with (W) and without (W/O) anomalous pixels in training datasets, which are labeled as OTCMA-W, OTCMA-W/O, CAE-W, and CAE-W/O. We fix the learning rate as 0.0001 for all experiments for fair comparison. To train the model without utilizing anomaly pixels, we eliminate all spectral pixels of anomaly according to the ground truth. The experiments are conducted on San Diego (San Diego-I and San Diego-II) dataset and HYDICE dataset, the false color image and its corresponding ground truth are presented in the final line of Fig. 9 (a) and (b). The comparative results on these three datasets are shown in Table IV and Fig. 9, respectively.

First, OTCMA with memory module shows significant improvements on both San Diego and HYDICE datasets. In the case of that train the model mixed with anomalies, our OTCMA-W increases the results of CAE-W on the San Diego-I, San Diego-II, and HYDICE datasets by 1.28%, 1.82%, and 0.04%. In particular, on the San Diego dataset, the results of OTCMA-W are even better than CAE-W/O trained without anomalies. These results prove the effectiveness of our memory module, which can effectively suppress the reconstruction ability of anomalies. Intrinsically, the AE is merely trained to reconstruct the background pixels based on their corresponding memory prototypes, which are semantic cluster centers of different kinds of land covers in a scene. Memory prototypes are more likely to be pure-spectral pixels. Instead of directly inputting the feature to the decoder, our OTCMA first read corresponding features from memory, which is the weighted summation of the memory prototypes with respect to their cluster predictions of current samples. Intuitively, similar spectral pixels will get consistent features after passing the memory module. So, the reconstruction ability of the AE is restricted only well to reconstruct these related features of pure memory prototypes (the cluster centers). This operation can effectively prevent the anomaly from being reconstructed. Although some memory prototypes correspond to the anomaly, they are difficult to play a destructive role, because anomaly-relevant memory prototypes are the minority and will be averaged by weighted reading operation. Therefore, as shown in Fig. 9(e) and (f), the anomalies detected by our method also reflect a strong anomaly score even on the edges of the objects.

Second, in eliminating anomaly pixels in the training process, OTCMA-W/O also performs better than CAE-W/O. The results (CAE-W, CAE-W/O) shown in Table IV prove that even if all anomalous pixels are eliminated from the training dataset, the performance improvement of CAE is not apparent. Especially, for the San Diego-II dataset, only 0.03% growth occurred after removing the anomalous pixels. The reason is that some mixed-spectral pixels contain anomaly characteristics. These mixed-spectral pixels usually exist on the edges of an object in the scene, for example, the edges of the airplane in the San Diego dataset. As shown in Fig. 9(c), we can see that the anomaly response of these pixels is weaker than the pixels inside the edges. Therefore, the trained CAE model can still reconstruct anomaly spectral pixels. By contrast, our proposed memory module can reduce the negative influence of mixed-spectral pixels and achieve better AD performance than CAE. CAE shows good performance when the anomalous object is tiny in the HYDICE dataset shown in Fig. 9(c). CAE-W has got 99.70% AUROC value in the HYDICE dataset and increased to 99.81% after removing the anomalies in the training process. Compared with the San Diego dataset, the anomalous pixels in the HYDICE dataset can be almost neglected, and they are isolated in the scene, which perfectly follows the assumption that anomaly pixels are the minority that deviated significantly from the background. So, CAE shows ideal results on the HYDICE dataset. In contrast to CAE, our proposed method OTCMA achieves significant results on both large and small anomalies.

In summary, OTCMA obtains better performance than CAE under different training settings (with or without anomalous

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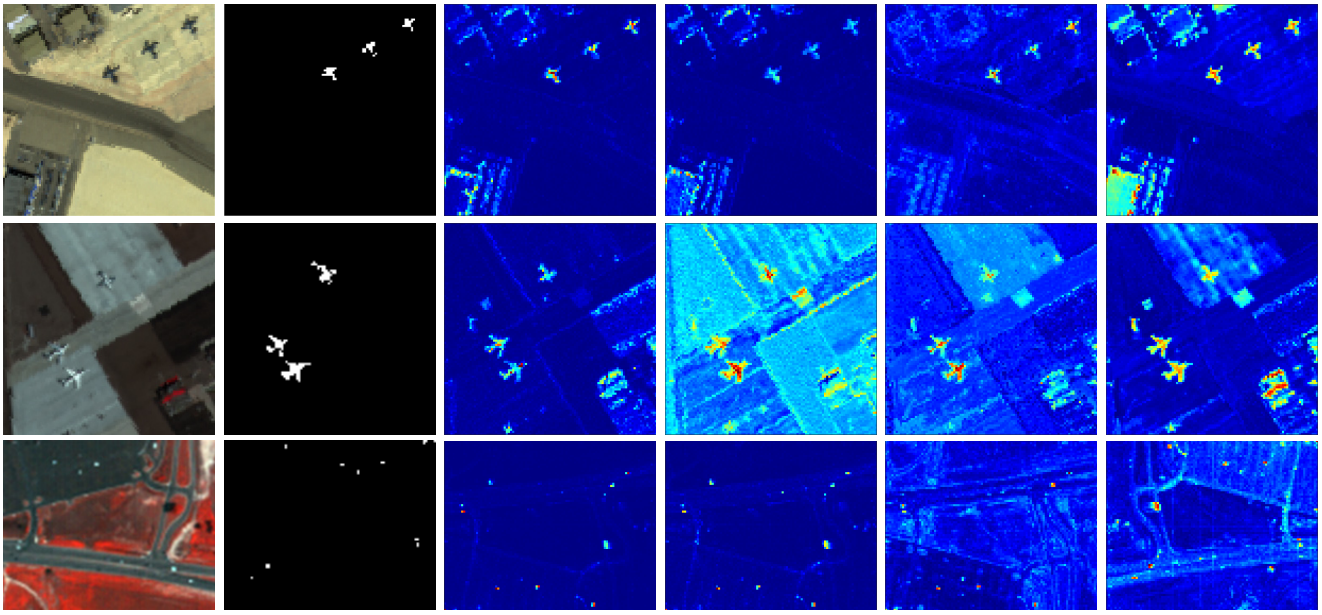


Fig. 9. Image descriptions and detection maps of San Diego-I, San Diego-II, and HYDICE datasets. (a) False color images. (b) Corresponding ground truth. (c) Detection maps of CAE trained without anomalous pixels. (d) Detection maps of CAE trained with anomalous pixels. (e) Detection maps of OTCMA trained without anomalous pixels. (f) Detection maps of OTCMA trained with anomalous pixels.

pixels in the training dataset), demonstrating the memory module's effectiveness.

2) Reconstructed Spectrum and T-SNE Results Analysis: To further verify the effectiveness of our memory module, we show the reconstructed spectrum of background and anomaly, respectively. Fig. 10 presents the spectral curves of original inputs and the reconstruction curves from CAE and OTCMA training without anomaly.

Compared with the visualized results in the first column (original input spectral curves) and the third column (reconstructed spectral curves by CAE) in Fig. 10, we can see that the original spectral curves and the reconstructed spectral curves of background and anomaly are similar. While the anomalous spectral curves are reconstructed into the background by OTCMA, please refer to the second column in Fig. 10. There are significant differences between the original spectrum and the reconstructed spectrum of anomaly. Meanwhile, OTCMA can keep a high similarity between the original spectrum and the reconstructed spectrum of the background. These results provide a compelling explanation for the effectiveness of our proposed memory module.

It should be noted that the discrimination of features is another key factor in the success of our model. Discriminative features effectively reduce the redundancy of the learned feature, such that they can only represent themselves. Besides, the features of the same pixels will concentrate together, as shown in Fig. 10(d) and (i). Compared with the features learned from CAE-W/O [Fig. 10(e) and (j)], our method can effectively push the anomaly pixels away from the background. Due to the significant difference between anomaly and background pixels in the feature space, the anomaly can be effectively detected. This is why Mahalanobis distance can work as an anomaly score in our model.

3) What We Learn in the Memory Module: As seen from the abovementioned discussion, the memory module plays a crucial and positive role, and it can boost HSI-AD performances by a fairly large margin. To explore what we learn in the memory module, we intuitively present the memory prototypes in the memory bank.

As San Diego-I dataset for example, we first train our OTCMA model using five memory prototypes. Then, we input memory prototypes to the decoder and acquire the corresponding reconstructed spectral curves, as shown in Fig. 11. The memory prototypes can be viewed as the cluster centers of different land covers learned by the OT process. From Fig. 11, we can see that there are some differences between these reconstructed spectral curves, which represent different background materials in the scene. So, the complex and various background pixels can be reconstructed well based on the memory module. In addition, please compare the spectral curves in Figs. 10(a) and 11, and it can be seen that the reconstructed spectral curves from memory prototypes have high similarity with the average spectrum of background pixels but significant differences with the average spectrum of anomaly. Therefore, the memory module can effectively mitigate the negative effects of the mixed anomalous pixels in the training process.

4) Analysis of the Number of Memory Prototypes: To discuss the effects of the number of memory prototypes, we set the number of memory prototypes $K = [3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25]$. We conducted the experiments on the San Diego-I, San Diego-II, and HYDICE datasets, and the results are shown in Fig. 12. When the number of the memory prototypes increases from 3 to 5, there are growths in these three datasets. After that, the

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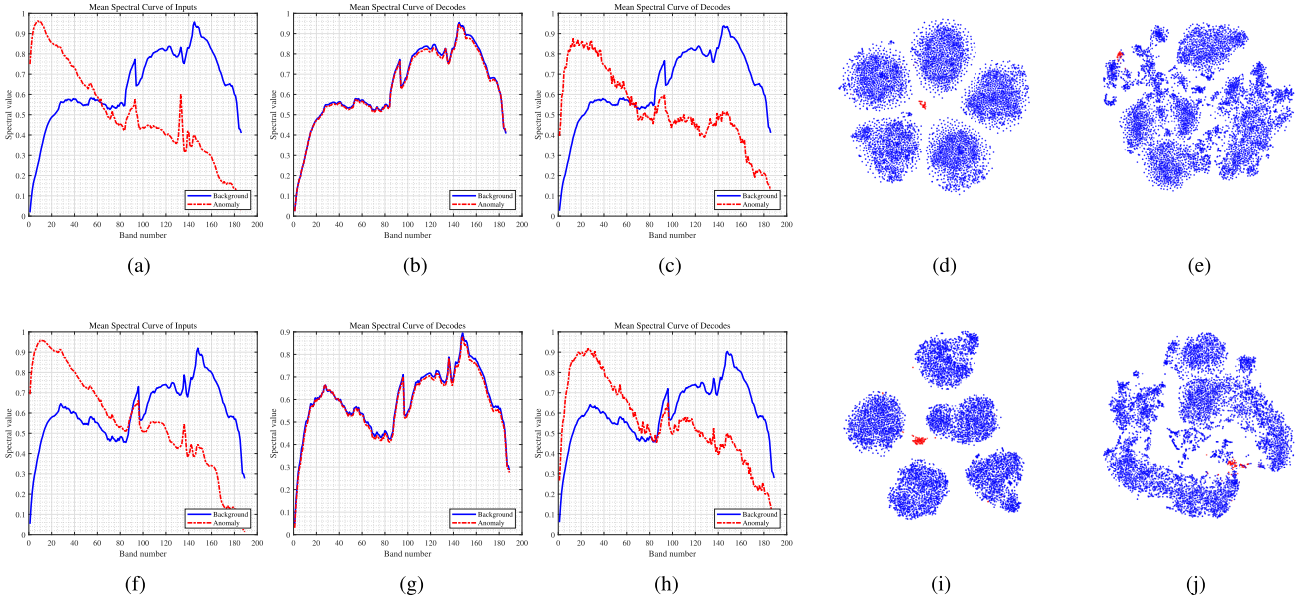


Fig. 10. Comparison of the spectral curves and T-SNE visualization obtaining from CAE-W/O and our proposed OTCMA-W/O. The blue solid lines and red dashed lines represent the average spectral curves of background and anomaly. The first row shows the results of the San Diego-I dataset, and the second row shows the results of the San Diego-II dataset. The first column presents the original spectral curves, the second column shows the corresponding reconstructed curves produced by OTCMA-W/O, the third column presents the reconstructed curves obtained from CAE-W/O (our model without memory module), the fourth column represents the T-SNE results of the encodes of OTCMA-W/O, and the fifth column shows the corresponding results from CAE-W/O. The blue dots are background, and the red ones are the anomaly. (a) San Diego-I. (b) OTCMA San Diego-I. (c) CAE San Diego-I. (d) OTCMA San Diego-I. (e) CAE San Diego-I. (f) San Diego-II. (g) OTCMA San Diego-II. (h) CAE San Diego-II. (i) OTCMA San Diego-II. (j) CAE San Diego-II.

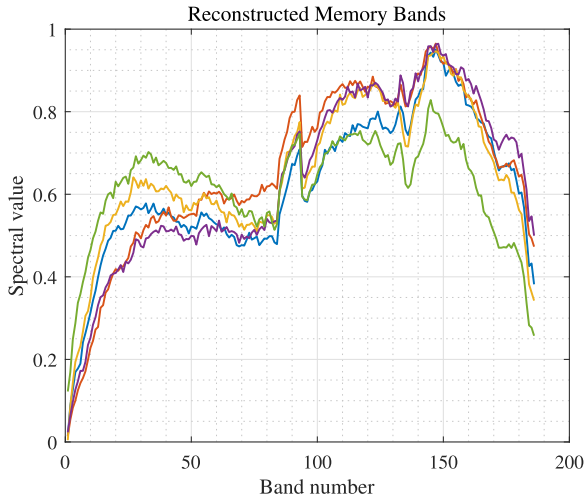


Fig. 11. Spectral curves of memory prototypes on the San Diego-I dataset.

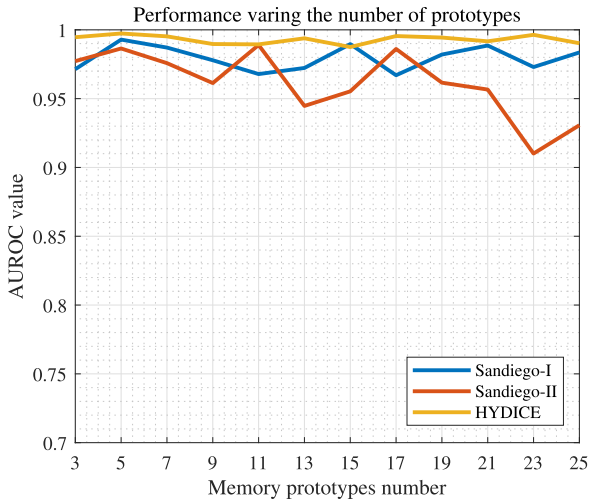


Fig. 12. AUROC values obtained by varying the number of memory prototypes from 3 to 25.

performance begins to decline with fluctuation. When the number of memory prototypes increases, the case becomes complicated. For the HYDICE dataset, OTCMA can be more stable in this dataset, because the anomalies in this scene are isolated pixels. They account for a very small proportion of the whole image. Even if the number of prototypes increases, the anomaly pixels are difficult to act as prototypes in the memory module. Therefore, the anomaly cannot be reconstructed. For the San Diego-I and San Diego-II datasets, the AUROC values fluctuated between 0.91 and 1, while the memory prototypes number varies from 3 to 25. In these datasets, the size of anomalies is relatively larger. More prototypes mean that memory is more likely to contain prototypes about anomalies. As a result, the AUROC values fluctuate and tend to decline when the prototype number increases. Considering the abovementioned reasons, we set the number of prototypes to be smaller, i.e., $K = 5$ for all the experiments. By alternative learning better features using pseudo-labels and estimating more reliable pseudo-labels according to the newly trained features, initialized pseudo-labels can be continually corrected in the training procedure. The model can get a stable clustering result after training.

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TABLE V
RUNNING TIMES OF EACH METHOD FOR THE ABU DATASET

Dataset	Global-RX	Local-RX	SVDD	CRD	AED	CAE	HADGAN	AEAN	MVSkt	OTCMA-E
Average test time	0.08s	38.86s	14.27s	35.93s	0.80s	3.4s	0.30s	25.33s	7.20s	3.75s

F. Comparisons of Computation Times

This section reports the average computation time of the methods on the ABU dataset. Global-RX, Local-RX, CRD, SVDD, and AED are implemented on Intel Xeon Processor E5-2630 v3 2.40 GHz $\times$ 20. CAE, HADGAN, AEAN, and MVSkt are implemented on Intel Xeon Processor E5-2630 v3 2.40 GHz $\times$ 20 and NVIDIA GTX1080. We adopt the optimal dual window size for Local-RX and CRD. In general, the deep learning-based methods have longer training times than the classical methods, so we only present the computation time of detection. The computation times are given in Table V.

The results demonstrate that our methods can get a considerable computation time compared with AEAN and MVSkt and have nearly the same computation times as CAE. HADGAN achieves the best computation time owing to its simple network. CAE shares the same encoder and the decoder with OTCMA-E, so they get almost the same computation times, and the slightly longer computations are cost in calculating the Mahalanobis distance and the ensemble of different anomaly measurements.

V. CONCLUSION

This article proposes a cluster-memory augmented deep AE via optimal transportation (OTCMA) for hyperspectral AD (HSI-AD). By introducing the deep clustering based on OT, the learned features can be more compacted (consistent and discriminative feature learning), so that they cannot represent anomalies. We propose a memory module to reconstruct a given spectral pixel indirectly, and this operation further reduces the possibility that anomalies being well reconstructed. Meanwhile, the memory module can effectively avoid the anomalies' interference mixed in the training dataset. Experimental results on the benchmark datasets, which have various data distributions, have demonstrated the superiority of our OTCMA compared with both advanced traditional and deep learning-based HSI-AD methods. In addition, the detailed analysis in the experiments shows that our proposed cluster-based memory module can effectively enhance the consistency and discriminability of the features and yield better performance than the other methods. Moreover, we have shown that spatial information can boost AD performance further. In the future, we will study the joint learning of spatial and spectral features for HSI-AD.

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